CC2541 Mini Development Kit Quick Start Guide

User's Guide

Literature Number: SWRU332C
November 2012 – Revised October 2016

Contents

1 Opening the Box and Evaluating Bluetooth® Low Energy 4
- 1.1 Kit Contents 4
- 1.2 Introduction 5
- 1.3 Hardware Setup 5

2 Evaluate Using BTool 6
- 2.1 Download and Install BLE-Stack 6
- 2.2 Identify the COM Port Number 6
- 2.3 Start BTool 6
- 2.4 Connect to the CC2541 Keyfob 6
- 2.5 Button Notifications 7
- 2.6 Enable Accelerometer 8
- 2.7 Immediate Alert 8
- 2.8 Terminate Connection 9
- 2.9 Source Code 9

3 Evaluate Using an iOS Device (iPod, iPad, or iPhone) 10
- 3.1 Download the SimpleLink™ Starter 10
- 3.2 Connect to the CC2541 Keyfob 11
- 3.3 Evaluate the Application 12

4 Additional Tools and Links 13
- 4.1 TI Packet Sniffer 13
- 4.2 SmartRF Flash Programmer 13
- 4.3 IAR Embedded Workbench 14
- 4.4 BLE E2E Forum 14
- 4.5 BLE Wiki 15
- 4.6 Useful Links 15

Revision History 15

Copyright © 2012–2016, Texas Instruments Incorporated

List of Figures

1 Insert Battery 5
2 Port Settings 6
3 Scan Button 7
4 Establish Button 7
5 Characteristic Value Handle 7
6 Discover UUIDs 8
7 Terminate Button 9
8 SimpleLink™ Starter 10
9 Connect to the Keyfob 11
10 Evaluate App 12
11 BLE Packet Sniffer 13
12 SmartRF Flash Programmer 13
13 IAR Embedded Workbench 14
14 BLE E2E Forum 14

SimpleLink is a trademark of Texas Instruments.
Bluetooth is a registered trademark of Bluetooth SIG Inc..
iOS is a trademark of Cisco.
Windows is a registered trademark of Microsoft Corp..

Copyright © 2012–2016, Texas Instruments Incorporated

CC2541 Mini Development Kit Quick Start Guide

1 Opening the Box and Evaluating Bluetooth® Low Energy

1.1 Kit Contents

- 1 × CC2540 USB dongle
- 1 × CC2541 Keyfob board
- 1 × Keyfob plastic case
- 1 × CC Debugger with cables
- 1 × CR2032 Battery
- Documentation

The RF boards in this kit are designed to comply with ETIS, FCC, and IC regulations over a temperature from 0°C to +35°C. The kit is for evaluation only; it is not FCC approved for resale.

CAUTION

The kit contains ESD sensitive components. Handle with care to prevent permanent damage.

Copyright © 2012–2016, Texas Instruments Incorporated

1.2 Introduction

This document is a guide through the initial steps required to run the preprogrammed Bluetooth® low energy (BLE) Keyfob demo application.

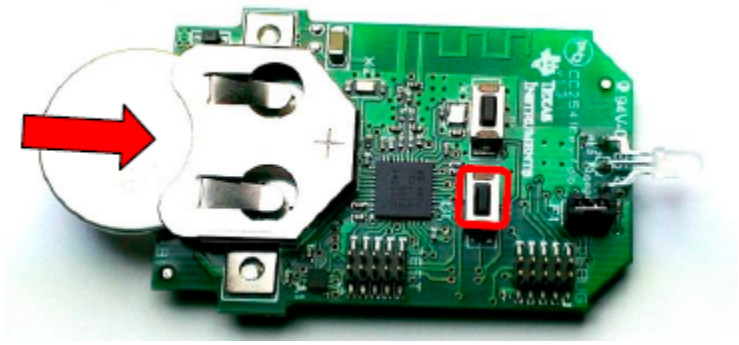
This guide describes the hardware in the box and some of the tools that can be used for developing software at a later stage. For the CC2541DK-MINI, there are two ways of getting started:

1. Evaluate using BTool: BTool is a Windows® application that allows the user to control a central device using the serial interface, and perform various BLE functions while connected to a peripheral device, such as the CC2541 Keyfob.
2. Evaluate using an iOS™ device: Some Apple iOS devices support BT4.0, and Texas Instruments has created an iOS application to evaluate a peripheral device, such as the CC2541 Keyfob. The iOS application runs on:
   - iPhone 4s / 5 / 5s / 5c / 6 / 6Plus / 6s
   - iPad 3 / 4 / air / air 2
   - iPod Touch (5. gen)

1.3 Hardware Setup

First, power up the CC2541 Keyfob. Insert the CR 2032 battery, as shown in Figure 1; the LED is lit green for one second.

Figure 1. Insert Battery



Toggle advertisements on and off by pushing the right button on the CC2541 Keyfob. During advertisement, the LED blinks red.

WARNING

This kit includes a non-rechargeable lithium battery. To minimize risk of personal injury or property damage due to a potential explosion or rupture of the battery, always completely remove the battery from the CC2541 Keyfob when connected to an external power source. The external power source cannot exceed 3.6 VDC. Dispose of the battery properly, and keep out of the reach of children at all times.

Copyright © 2012–2016, Texas Instruments Incorporated

2 Evaluate Using BTool

2.1 Download and Install BLE-Stack

The latest BLE software can be downloaded at www.ti.com/ble-stack.

After the BLE-Stack software installation is complete, the USB dongle driver must be associated with the device to use the BTool application. To associate the USB dongle driver, first connect the USB dongle to the USB port of the PC, or to a USB hub that connects to the PC.

The first time the dongle is connected to the PC, a message pops up indicating that Windows does not recognize the device.

The driver is found in the Accessories\ Drivers folder in the default install directory. For more information on how to install the driver, refer to the *CC2540/41 Mini Development Kit User Guide* (SWRU270).

2.2 Identify the COM Port Number

When the driver is installed, determine which COM port Windows has assigned to the USB dongle. Right-click on Computer in the Start Menu, and select Properties.

The System Properties window opens; select Device Manager.

A list of all hardware devices appears. Under the Ports (COM & LPT) section, the device TI CC2540 Low-Power RF to USB CDC Serial Port appears. Next to the name should be the port number (COM#).

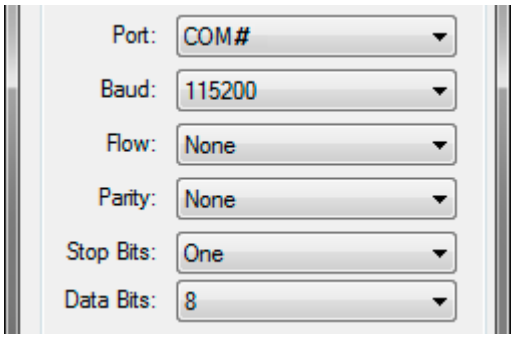
Note this port number, as it is needed to use BTool.

2.3 Start BTool

BTool is included as part of the installation of the BLE stack, and can be found in the \Projects\BTool folder in the default install directory.

When starting up BTool, you will be prompted to set port settings. Select the options shown in Figure 2, and press OK.

Figure 2. Port Settings

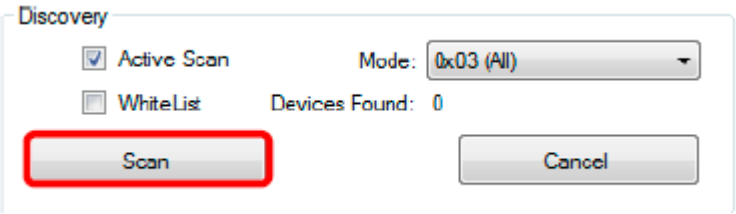


2.4 Connect to the CC2541 Keyfob

Pressing the right button on the CC2541 Keyfob starts the advertisement. The device advertises for 30 seconds. In BTool, press the Scan button, as shown in Figure 3.

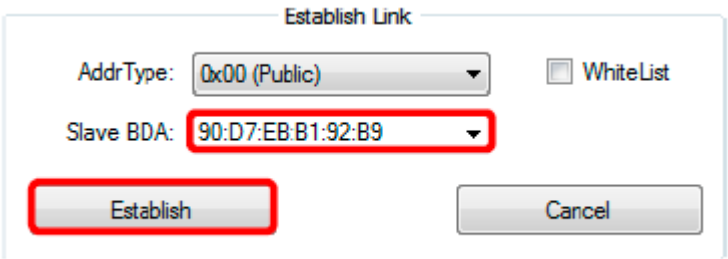
Copyright © 2012–2016, Texas Instruments Incorporated

Figure 3. Scan Button



After the scanning is complete, choose the device to connect to, and press Establish, as shown in Figure 4.

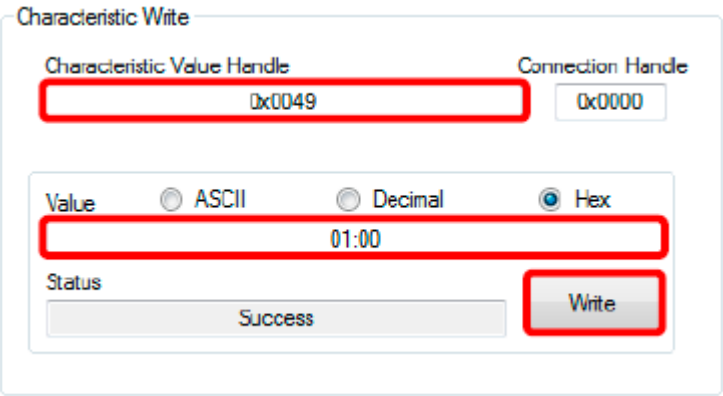
Figure 4. Establish Button



2.5 Button Notifications

To notify when buttons are pressed on the CC2541 Keyfob, notifications must be enabled. This is done in the Read/Write tab of BTool by writing 01:00 to the characteristic handle 0x0049, as shown in Figure 5.

Figure 5. Characteristic Value Handle

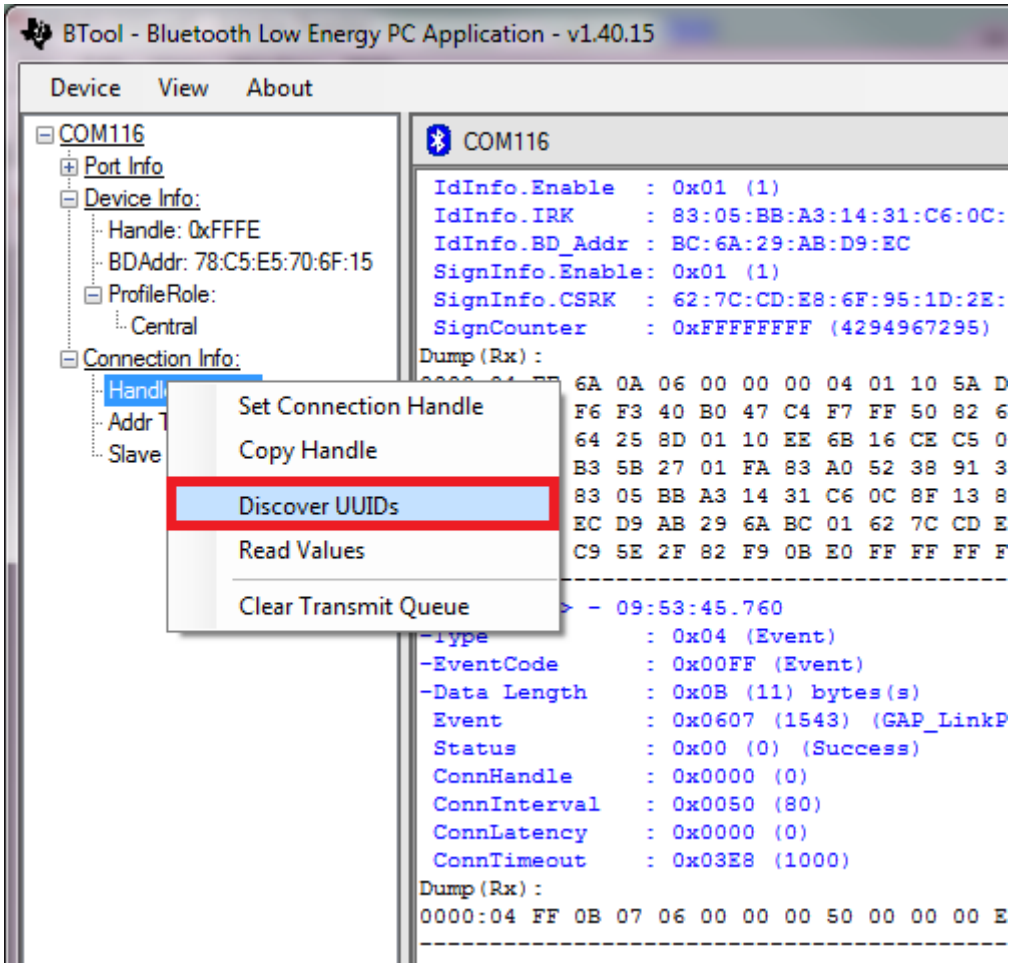


If a button on the CC2541 Keyfob is now pushed, notifications are sent, and can be monitored in the BTool log window.

NOTE: These handles are subject to change, depending on the firmware version used. To verify handle IDs, discover all UUIDs by right-clicking the handle id under Connection Info, and selecting Discover UUIDs, as shown in Figure 6.

Copyright © 2012–2016, Texas Instruments Incorporated

Figure 6. Discover UUIDs



2.6 Enable Accelerometer

Similar to the button notifications, the notifications for the accelerometer data can be enabled. This example only shows enable notifications for the x-axis.

First, enable the accelerometer by writing 01:00 to the characteristic handle 0x0035 in the Read/Write tab of BTool. To enable notifications for the xaxis, write 01:00 to the characteristic handle 0x003C. This enables the CC2541 Keyfob to send notifications as it is moved.

For more information about the accelerometer service, refer to the *CC2540/41 Mini Development Kit User Guide* (SWRU270).

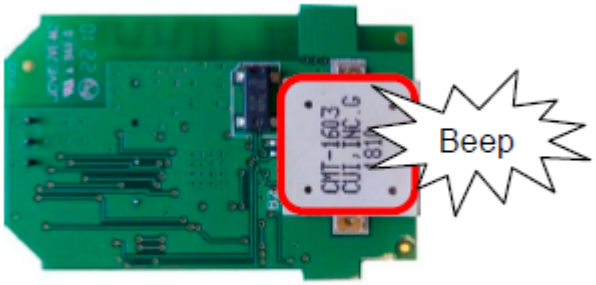
2.7 Immediate Alert

To sound the buzzer located on the CC2541 Keyfob, write the following value to the characteristic handle 0x0028:

Copyright © 2012–2016, Texas Instruments Incorporated

- 01:00 for low Alert
- 02:00 for high Alert
- 00:00 to turn off

The buzzer will sound for 10 seconds.

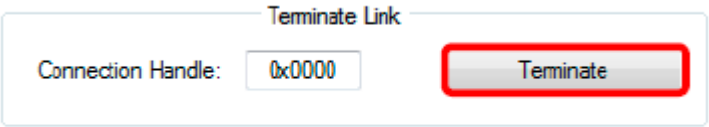


2.8 Terminate Connection

There are three options to terminate the connection:

- Press the Terminate button in BTool, as shown in Figure 7.
- Remove the battery from the CC2541 Keyfob, which triggers a supervision timeout.
- Move the CC2541 Keyfob out of range (typically >10m), which triggers a supervision timeout.

Figure 7. Terminate Button



2.9 Source Code

The project and source code files for the preprogrammed application (as well as many others) are included with the Bluetooth low energy (BLE) stack from Texas Instruments, which can be downloaded at www.ti.com/ble-stack.

The project implementing this demo is called Keyfobdemo (CC2541DK-mini Keyfob Slave configuration). The project can be modified as desired, and should provide a good framework for developing custom BLE applications.

More details on these projects can be found within the *BLE Software Developer's Guide* (SWRU271), which is also included with the stack installer. For troubleshooting, refer to the *CC2540/41 Mini Development Kit User Guide* (SWRU270).

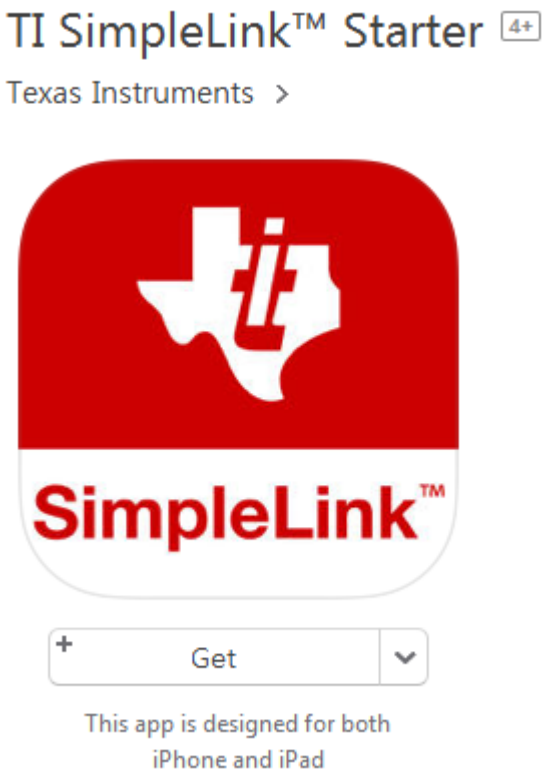
Copyright © 2012–2016, Texas Instruments Incorporated

3 Evaluate Using an iOS Device (iPod, iPad, or iPhone)

3.1 Download the SimpleLink™ Starter

The SimpleLink™ Starter iOS app can be downloaded through iTunes (found at www.apple.com/itunes) or the App Store, which is pre-installed on iOS devices.

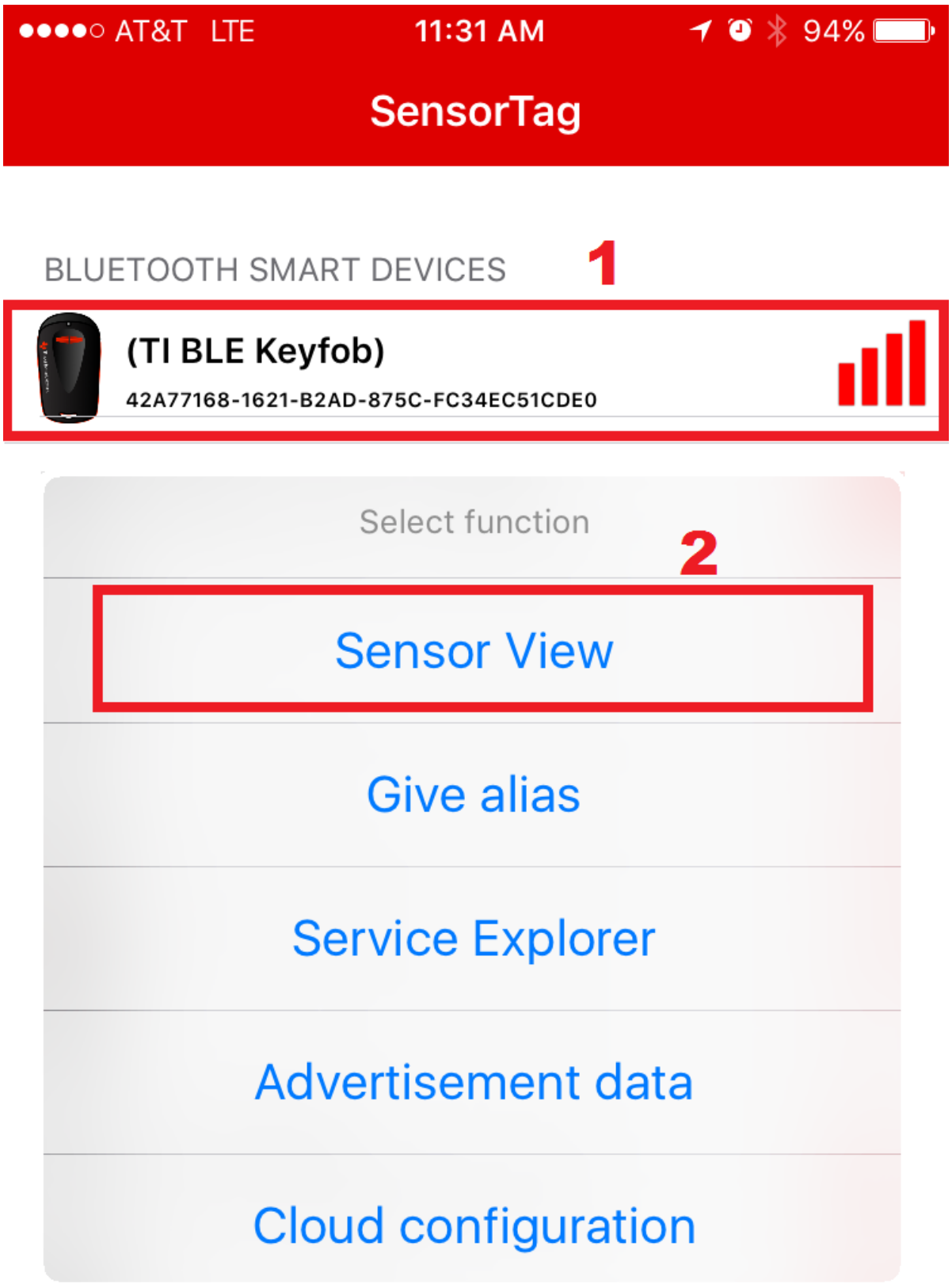
Figure 8. SimpleLink™ Starter



Copyright © 2012–2016, Texas Instruments Incorporated

3.2 Connect to the CC2541 Keyfob

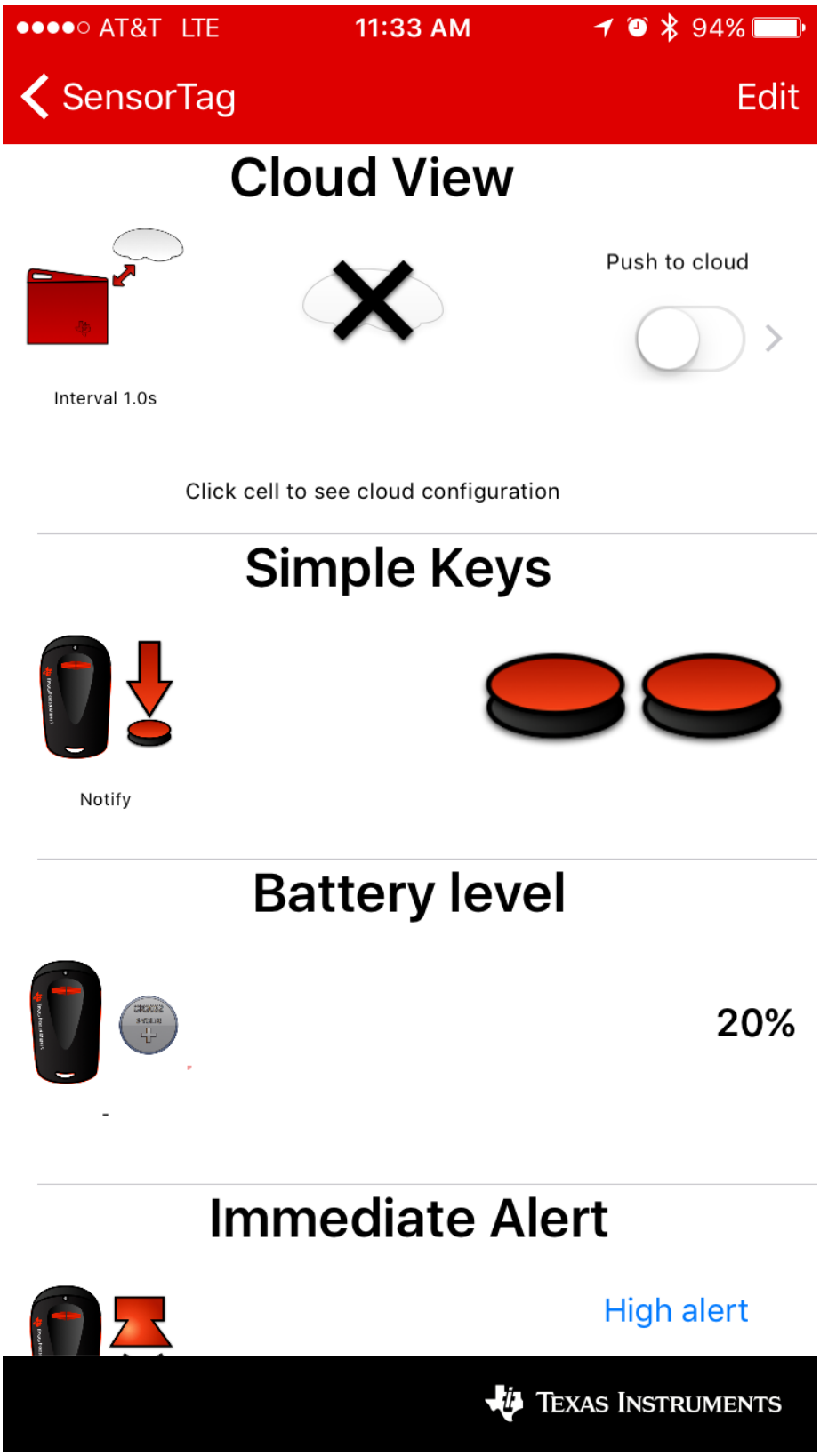
Figure 9. Connect to the Keyfob



Copyright © 2012–2016, Texas Instruments Incorporated

3.3 Evaluate the Application

Figure 10. Evaluate App



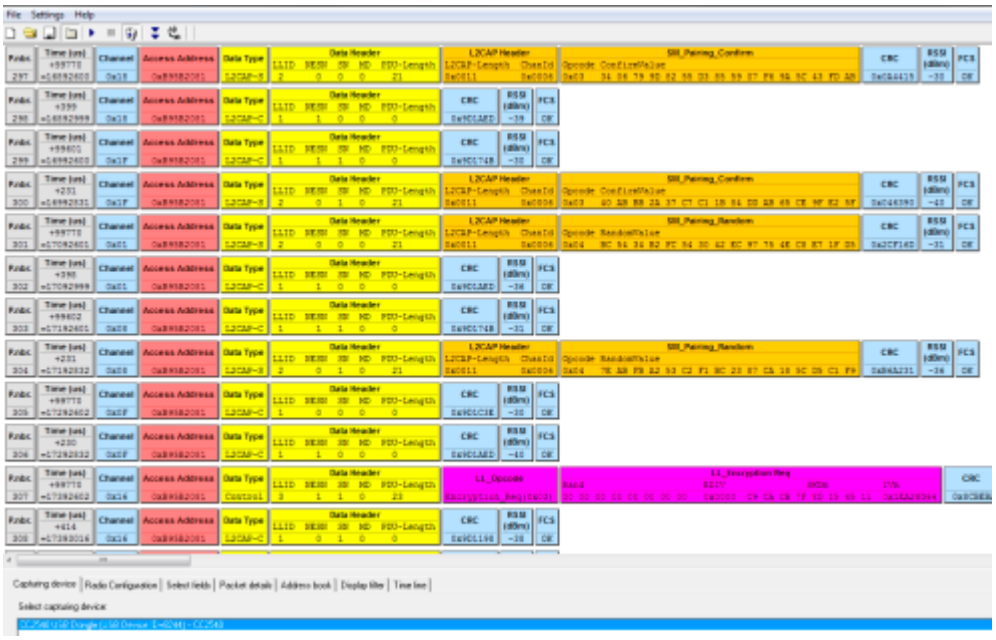
Copyright © 2012–2016, Texas Instruments Incorporated

4 Additional Tools and Links

4.1 TI Packet Sniffer

The CC2540 USB dongle loaded with the TI Packet Sniffer firmware can be used as a BLE sniffer, and monitor packets while the iPhone 4S Demo is running.

Figure 11. BLE Packet Sniffer

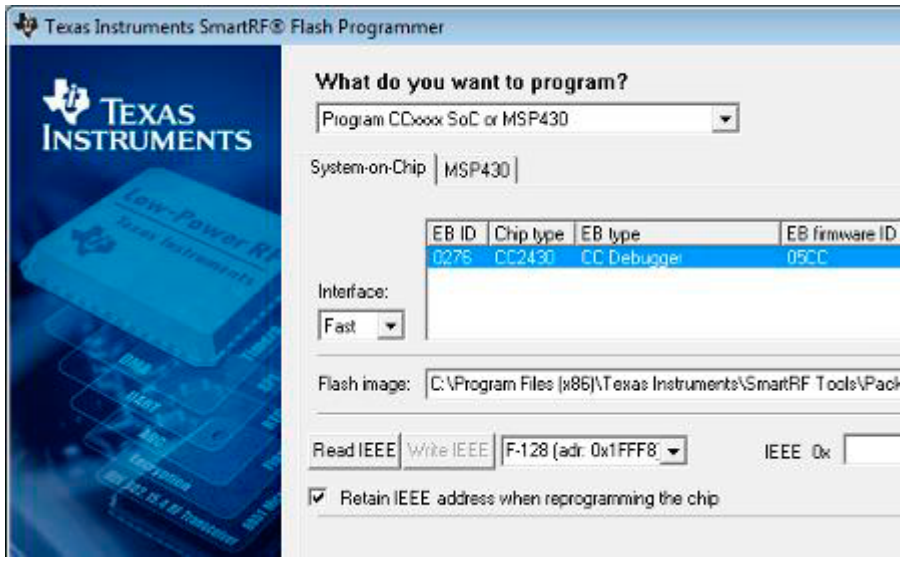


The TI Packet Sniffer software can be downloaded at www.ti.com/tool/packet-sniffer.

4.2 SmartRF Flash Programmer

Texas Instruments has a simple tool to program and flash the CC2541.

Figure 12. SmartRF Flash Programmer



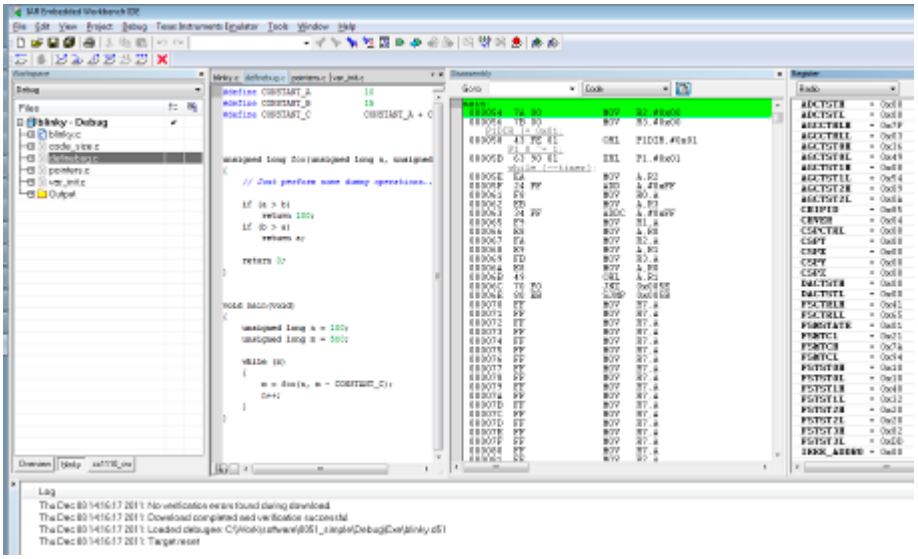
SmartRF Flash Programmer can be downloaded at www.ti.com/tool/flash-programmer.

Copyright © 2012–2016, Texas Instruments Incorporated

4.3 IAR Embedded Workbench

To develop software, program, and debug the CC2541, use the IAR Embedded Workbench for 8051.

Figure 13. IAR Embedded Workbench

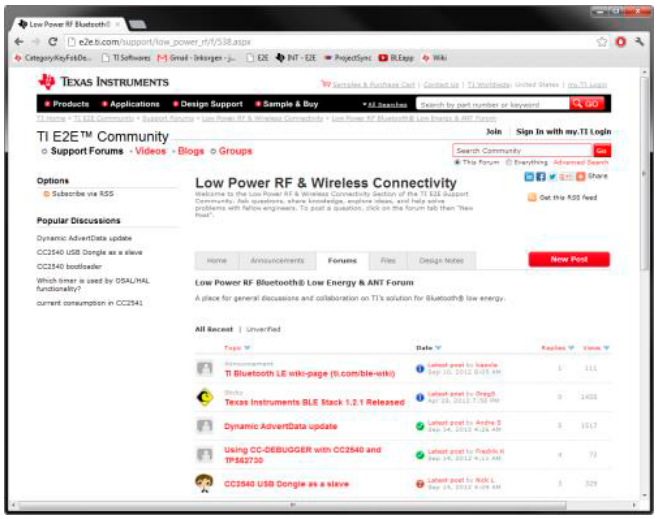


More information on IAR EW8051, including a free evaluation version download, can be found at https://www.iar.com/iar-embedded-workbench/partners/texas-instruments/ti-wireless/. Refer to the *BLE Software Developer's Guide* (SWRU271) for the specific version of IAR.

4.4 BLE E2E Forum

For additional help, visit the TI Bluetooth low energy E2E forum, www.ti.com/ble-forum, for instant support during development.

Figure 14. BLE E2E Forum



Copyright © 2012–2016, Texas Instruments Incorporated

4.5 BLE Wiki

The BLE Wiki contains application examples, guides, and documentation covering any extra steps you might need help with. The Wiki is not only managed by Texas Instruments employees, but also by E2E community members. Anyone can share, edit, and make use of the information posted here.

The Wiki is found at www.ti.com/ble-wiki.

4.6 Useful Links

- TI BLE Stack and Software: www.ti.com/ble-stack
- *CC2540/41 Mini Development Kit User Guide*: www.ti.com/lit/swru270
- *CC2540/41 BLE Software Developer's Guide*: www.ti.com/lit/swru271
- *CC2540/41 User's Guide*: www.ti.com/lit/swru191
- CC2541 Product Page: www.ti.com/product/cc2541

Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from B Revision (October 2015) to C Revision	Page
• Changed Characteristic Handle from 0x0048 to 0x0049.	7
• Updated Characteristic Value Handle image.	7
• Changed Characteristic Handle from 0x0034 to 0x0035.	8
• Changed Characteristic Handle from 0x003B to 0x003C.	8

Copyright © 2012–2016, Texas Instruments Incorporated

IMPORTANT NOTICE

Texas Instruments Incorporated and its subsidiaries (TI) reserve the right to make corrections, enhancements, improvements and other changes to its semiconductor products and services per JESD46, latest issue, and to discontinue any product or service per JESD48, latest issue. Buyers should obtain the latest relevant information before placing orders and should verify that such information is current and complete. All semiconductor products (also referred to herein as “components”) are sold subject to TI’s terms and conditions of sale supplied at the time of order acknowledgment.

TI warrants performance of its components to the specifications applicable at the time of sale, in accordance with the warranty in TI’s terms and conditions of sale of semiconductor products. Testing and other quality control techniques are used to the extent TI deems necessary to support this warranty. Except where mandated by applicable law, testing of all parameters of each component is not necessarily performed.

TI assumes no liability for applications assistance or the design of Buyers’ products. Buyers are responsible for their products and applications using TI components. To minimize the risks associated with Buyers’ products and applications, Buyers should provide adequate design and operating safeguards.

TI does not warrant or represent that any license, either express or implied, is granted under any patent right, copyright, mask work right, or other intellectual property right relating to any combination, machine, or process in which TI components or services are used. Information published by TI regarding third-party products or services does not constitute a license to use such products or services or a warranty or endorsement thereof. Use of such information may require a license from a third party under the patents or other intellectual property of the third party, or a license from TI under the patents or other intellectual property of TI.

Reproduction of significant portions of TI information in TI data books or data sheets is permissible only if reproduction is without alteration and is accompanied by all associated warranties, conditions, limitations, and notices. TI is not responsible or liable for such altered documentation. Information of third parties may be subject to additional restrictions.

Resale of TI components or services with statements different from or beyond the parameters stated by TI for that component or service voids all express and any implied warranties for the associated TI component or service and is an unfair and deceptive business practice. TI is not responsible or liable for any such statements.

Buyer acknowledges and agrees that it is solely responsible for compliance with all legal, regulatory and safety-related requirements concerning its products, and any use of TI components in its applications, notwithstanding any applications-related information or support that may be provided by TI. Buyer represents and agrees that it has all the necessary expertise to create and implement safeguards which anticipate dangerous consequences of failures, monitor failures and their consequences, lessen the likelihood of failures that might cause harm and take appropriate remedial actions. Buyer will fully indemnify TI and its representatives against any damages arising out of the use of any TI components in safety-critical applications.

In some cases, TI components may be promoted specifically to facilitate safety-related applications. With such components, TI’s goal is to help enable customers to design and create their own end-product solutions that meet applicable functional safety standards and requirements. Nonetheless, such components are subject to these terms.

No TI components are authorized for use in FDA Class III (or similar life-critical medical equipment) unless authorized officers of the parties have executed a special agreement specifically governing such use.

Only those TI components which TI has specifically designated as military grade or “enhanced plastic” are designed and intended for use in military/aerospace applications or environments. Buyer acknowledges and agrees that any military or aerospace use of TI components which have ***not*** been so designated is solely at the Buyer's risk, and that Buyer is solely responsible for compliance with all legal and regulatory requirements in connection with such use.

TI has specifically designated certain components as meeting ISO/TS16949 requirements, mainly for automotive use. In any case of use of non-designated products, TI will not be responsible for any failure to meet ISO/TS16949.

Products		**Applications**	
Audio	www.ti.com/audio	Automotive and Transportation	www.ti.com/automotive
Amplifiers	amplifier.ti.com	Communications and Telecom	www.ti.com/communications
Data Converters	dataconverter.ti.com	Computers and Peripherals	www.ti.com/computers
DLP® Products	www.dlp.com	Consumer Electronics	www.ti.com/consumer-apps
DSP	dsp.ti.com	Energy and Lighting	www.ti.com/energy
Clocks and Timers	www.ti.com/clocks	Industrial	www.ti.com/industrial
Interface	interface.ti.com	Medical	www.ti.com/medical
Logic	logic.ti.com	Security	www.ti.com/security
Power Mgmt	power.ti.com	Space, Avionics and Defense	www.ti.com/space-avionics-defense
Microcontrollers	microcontroller.ti.com	Video and Imaging	www.ti.com/video
RFID	www.ti-rfid.com		
OMAP Applications Processors	www.ti.com/omap	**TI E2E Community**	e2e.ti.com
Wireless Connectivity	www.ti.com/wirelessconnectivity		

Mailing Address: Texas Instruments, Post Office Box 655303, Dallas, Texas 75265
Copyright © 2016, Texas Instruments Incorporated